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$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$\text{var}\left(\frac{(n-1)s^2}{\sigma^2}\right) = 2(n-1) \Rightarrow \frac{(n-1)^2 \text{var}(s^2)}{\sigma^4} = 2(n-1)$$

$$\Rightarrow \text{var}(s^2) = \frac{2\sigma^4}{n-1}$$

Note:- Like normal distⁿ, CDF of a chi-square distⁿ is not in closed form. So, χ_n^2 distⁿ is tabulated for values of $n=1, 2, \dots$. Table usually go upto $n=30$. Since for $n > 30$, it is possible to use CLT.

Ques:- How to see a chi-square table?

Suppose we have $\chi_{n,\alpha}^2$ for the upper α -percentile of the χ_n^2 distⁿ, that is

$$P(\chi_n^2 > \chi_{n,\alpha}^2) = \alpha$$

So, using a table we can compute various prob. or quantiles for χ_n^2 distⁿ.

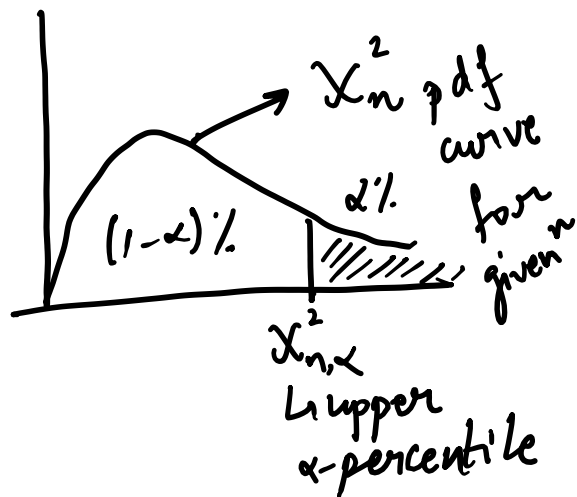
$$P(X \leq q_p) = p$$

50 percentile = 0.50 quantile

$q_{0.25}$ = 25% of prob. is below it

$$P(X > c) = \alpha \Rightarrow P(X \leq c) = 1 - \alpha \Rightarrow c = q_{1-\alpha}$$

$\Rightarrow$ an upper α percentile is same as $(1-\alpha)$ -quantile.



Ex:- $\chi_{0.05,10}^2$

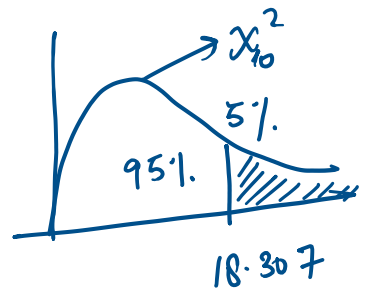
$\rightarrow 0.05$ = upper 0.05 percentile

10 = degree of freedom

$$\chi_{0.05,10}^2 = 18.307$$

Here $n=10$, we are looking for 5% upper percentile of a χ^2_{10} distⁿ

If $X \sim \chi^2_{10}$, $P(X > 18.307) = 0.05$



Ex:- $\chi^2_{0.025, 10}$ $\chi^2_{0.025, 15}$ $\chi^2_{0.01, 7}$ $\chi^2_{0.05, 24}$
 $\chi^2_{0.005, 10}$ $\chi^2_{0.005, 5}$

Soln:- 20.483, 27.488, 18.475, 36.415, 25.188, 16.750

$P(a < X \leq b) = P(X \leq b) - P(X \leq a)$

Ques:- $P(32.007 < X < \chi^2_{\alpha, 23}) = 0.09$
 $P(X > a) = 1 - P(X \leq a)$

Soln:- $P(X < \chi^2_{\alpha, 23}) - P(X \leq 32.007) = 0.09$

$\Rightarrow 1 - P(X \geq \chi^2_{\alpha, 23}) - 1 + P(X > 32.007) = 0.09$

$\Rightarrow -P(X \geq \chi^2_{\alpha, 23}) + 0.10 = 0.09$

$\Rightarrow P(X \geq \chi^2_{\alpha, 23}) = 0.01$

$\Rightarrow \alpha = 0.01$ and $\chi^2_{\alpha, 23} = 41.638$

Ques:- $P(\chi^2_{\alpha, 10} < X < 23.209) = 0.015$

Soln:- $\alpha = 0.025$, $\chi^2_{\alpha, 10} = 20.483$

Student's t-distribution

Let X and Y be two ind. RVs s.t. $X \sim N(0,1)$ and $Y \sim \chi_n^2$.

Then $T = \frac{X}{\sqrt{Y/n}}$ is said to have a student's t-distⁿ with n degree of freedom.

Denoted as $T \sim t_n$

The PDF of T is

$$f_T(t) = \frac{1}{\sqrt{n} B\left(\frac{n}{2}, \frac{1}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-\left(\frac{n+1}{2}\right)}, \quad -\infty < t < \infty$$

where $B(\cdot, \cdot)$ is beta fⁿ and defined as $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$

$$\Rightarrow f_T(t) = \frac{\frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}}}{\sqrt{\frac{n}{2}}} \left(1 + \frac{t^2}{n}\right)^{-\left(\frac{n+1}{2}\right)} \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\text{For } n=1, \quad f_T(t) = \frac{1}{\pi(1+t^2)} \quad -\infty < t < \infty$$

This is the standard Cauchy distⁿ.



Note:- PDF of t-distⁿ is sym. around $t=0$.

For symmetric, $P(T > t) = P(T < -t)$

$$F_T(-t) = 1 - F_T(t)$$

Proof:-

$$P(T > t)$$

$$\text{for } t > 0, \quad T = \frac{X}{\sqrt{Y/n}} \Rightarrow T > t \Rightarrow X > t \sqrt{\frac{Y}{n}}$$

$$T < -t \Rightarrow X < -t \sqrt{\frac{Y}{n}}$$

$$\therefore X \sim N(0,1) \Rightarrow P(X > a) = P(X < -a), \quad a > 0$$

Note:-

All odd order moments are zero. And even order is given by

$$E(T^k) = \frac{n^{k/2} \sqrt{\frac{k+1}{2}} \sqrt{\frac{n-k}{2}}}{\sqrt{\frac{1}{2}} \sqrt{\frac{n}{2}}}$$

Find mean and variance of t-distⁿ?

$$E(T) = 0 \quad (\text{first order moment})$$

$$\text{Var}(T) = E(T^2) - (E(T))^2 = E(T^2) = \frac{n}{n-2}, \quad n > 2$$

$$E(T^2) = \frac{n \sqrt{\frac{3}{2}} \sqrt{\frac{n-2}{2}}}{\sqrt{\frac{1}{2}} \sqrt{\frac{n}{2}}} = n \left(\frac{1}{2} \right) \frac{\sqrt{\frac{1}{2}} \sqrt{\frac{n}{2}}}{\sqrt{\frac{1}{2}} \left(\frac{n}{2} - 1 \right) \sqrt{\frac{1}{2} - 1}} = \frac{n}{n-2}, \quad n > 2$$

$\sqrt{n} = (n-1) \sqrt{n-1}$

Result:- Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Then

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$$

Proof:-

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2) \Rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$$

$$\text{and } (n-1) \frac{s^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$\bar{X} \sim N(\mu, \sigma^2/n) \Rightarrow \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0,1)$$

$$\Rightarrow \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0,1)$$

$$T = \frac{X}{\sqrt{Y/n}}$$

By the defⁿ of t -distⁿ,

$$\frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{\frac{(n-1)s^2}{\sigma^2(n-1)}}} \sim t_{n-1}$$

$$\Rightarrow \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{s/\sigma} = \frac{\sqrt{n}(\bar{X} - \mu)}{s} \sim t_{n-1}$$

$\Rightarrow$ This implies that t -distⁿ is a sampling distⁿ.

Claim:- Let $T \sim t_n$. As $n \rightarrow \infty$, the PDF of T converges to PDF of standard normal R.V.

Proof:- $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(0,1)$

$$X_n^2 = X_1^2 + X_2^2 + \dots + X_n^2$$

$$\Rightarrow \frac{X_n^2}{n} = \frac{1}{n} \sum X_i^2$$

$$E[X_i] = 1$$

$$T = \frac{X}{\sqrt{Y/n}}$$

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Law of large no's :- If Y_i are iid with finite mean μ ,

then $\frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{P} \mu$

Here $Y_i = X_i^2$ $\because E[X_i^2] = 1$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} 1$$

$$\Rightarrow \frac{X_n^2}{n} \xrightarrow{P} 1 \Rightarrow \sqrt{\frac{X_n^2}{n}} \xrightarrow{P} 1$$

$$T = \frac{X}{\sqrt{X_n^2/n}} \rightarrow X \sim N(0, 1)$$

PDF:- $f_T(t) = \frac{\sqrt{\frac{n+1}{2}}}{\sqrt{n\pi} \sqrt{\frac{n}{2}}} \underbrace{\left(1 + \frac{t^2}{n}\right)}_{\rightarrow 1}^{-\underbrace{\left(\frac{n+1}{2}\right)}_{\rightarrow \infty}}$